

Xukai Zhao

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RESEARCH PROFILE

My research explores how AI can reshape urban perception studies. I develop multimodal computational frameworks that integrate natural language processing, computer vision, and geospatial analysis to examine how people perceive and experience urban environments. By converting large-scale urban data into measurable and interpretable indicators, my work provides evidence for more responsive urban design, planning, and management.

EDUCATION

2025.09–	Ph.D. in Urban Planning , School of Architecture, Tsinghua University
2022.09–2025.06	Master of Landscape Architecture , South China University of Technology
2018.09–2022.06	Bachelor of Landscape Architecture , Soochow University
IELTS 7.5 (L 8.5 / R 8.5 / W 7.0 / S 6.5)	

PUBLICATIONS

Agentic Systems / RAG

- Under review** Xukai Zhao, et al. From Perceived Attractiveness to Visitation: Multi-Scale Measurement and Influencing Factors across 143 Chinese Historic Cities. *Tourism Management* (under 3rd review; Q1, IF 14.5)
A tourist-centered framework analyzing 610k reviews across 2,128 attractions, using an LLM pipeline to measure multi-scale perceived attractiveness and disentangle how attraction attributes and perceptions drive visitation.
- 2026.05 Xukai Zhao, et al. A multi-agent large language model workflow for analyzing perceived cultural values from social media: A study of 141 Chinese cities. *Cities* (Q1, IF 7.0)
Multi-agent LLM workflow that quantifies eight value dimensions from 284k reviews and constructs Cultural Heritage Perception Graphs and a RAG-based retrieval system to support evidence-based interpretation of public feedback.
- 2025.09 Yuxing Lu, Wei Wu, Xukai Zhao, et al. KARMA: Leveraging Multi-Agent LLMs for Automated Knowledge Graph Enrichment. *NeurIPS 2025 (Spotlight, Top 3%)*
Nine-agent LLM framework automating knowledge-graph enrichment from literature; on 1,200 PubMed articles it added 38k+ new entities at 83.1% verified correctness while cutting conflicting edges by 18.6%.
- 2025.09 Yuxing Lu, Gecheng Fu, Wei Wu, Xukai Zhao. Towards Doctor-Like Reasoning: Medical RAG Fusing Knowledge with Patient Analogy through Textual Gradients. *NeurIPS 2025 (Poster)*

LLM Benchmarks

- 2025.10 Yuxing Lu, Xukai Zhao, et al. MetaBench: A Multi-task Benchmark for Assessing LLMs in Metabolomics. *ACL 2026 (Oral, Top 3%)*
- 2025.07 Xukai Zhao, et al. Urban Planning in the Age of Large Language Models: Assessing OpenAI o1's Performance across 556 Tasks. *Computers, Environment and Urban Systems* (Q1, IF 8.9)
The first open benchmark for LLMs in urban planning, covering 556 tasks across documentation, professional exams, data analysis, algorithm support, and writing, revealing strengths and weaknesses of OpenAI o1.

Urban Perception (NLP & CV)

- 2025.03 Xukai Zhao, et al. Exploring Temporal-Spatial Patterns and Nonlinear Driving Mechanism of Park Perceptions: A Multi-Source Big Data Study. *Sustainable Cities and Society* (Q1, IF 13.3)
Measured perceived park accessibility, usability, and attractiveness across 270 parks, integrating perception scores with 2SfCA, XGBoost, SHAP, and PDP to reveal spatial temporal patterns and nonlinear drivers.
- 2025.02 Xukai Zhao, et al. Assessing and Interpreting Perceived Park Accessibility, Usability and Attractiveness through Texts and Images from Social Media. *Sustainable Cities and Society* (Q1, IF 13.3)
Developed a dual-modal park perception model using social media texts and images, achieving 86.81% accuracy and producing city-scale perception maps and SHAP-based interpretations for targeted park improvement.
- 2024.12 Wenwen Huang, Xukai Zhao, et al. How to Quantify Multidimensional Perception of Urban Parks?. *Urban Forestry & Urban Greening* (Q1, IF 8.2)
- 2024.07 Xukai Zhao, et al. Perception Evaluation of Urban Green Spaces Using Panoramic Images and Deep Learning. *Landscape Architecture Frontiers*
Built a panoramic green-space image library with crowd ratings and fine-tuned vision models (82.0% accuracy) plus semantic

segmentation and image classification for perception scoring.

- 2024.03 **Xukai Zhao**, et al. Multiscale Scoring Model for Enhanced Urban Perception Evaluation. *ICASSP 2024 (Poster)*
- 2023.12 **Xukai Zhao**, et al. An Integrated Deep Learning Approach for Assessing the Visual Qualities of Built Environments Utilizing Street View Images. *Engineering Applications of Artificial Intelligence (Q1, IF 9.0)*
Proposed a street view-based deep learning framework to predict six visual perceptions based on the Place Pulse 2.0 dataset, achieving 78.47% accuracy and using SHAP and CAM to explain perception drivers across Guangzhou.
- 2023.06 Yuxing Lu, **Xukai Zhao**, et al. Medical Knowledge-Enhanced Prompt Learning for Diagnosis Classification from Clinical Text. *ACL 2023 (Workshop)*

Other Co-Authored Work

- 2025.02 Wenke Wang, Yang Shi, Jie Zhang, **Xukai Zhao**. Exploration of Non-Linear Influence Mechanisms of Traditional Courtyard Forms on Thermal Comfort. *Sustainable Cities and Society (Q1, IF 13.3)*
- 2024.08 Yuxing Lu, **Xukai Zhao**. ClinicalRAG: Enhancing Clinical Decision Support through Heterogeneous Knowledge Retrieval. *ACL 2024 (Workshop)*
- 2022.04 Jiesi Wang, Guanting Zhang, **Xukai Zhao**. Evaluating the Effectiveness of Community Gardens by a Quantitative Systematic Framework. *Sustainable Cities and Society (Q1, IF 13.3)*

HONORS & SCHOLARSHIPS

- 2025 Tsinghua Future Scholar Scholarship
- 2024 National Scholarship for Graduate Students; He Jingtang Science & Technology Innovation Award
- 2023 Longhu Scholarship; He Jingtang Science & Technology Innovation Award

CONFERENCES & FORUMS

- 2023–2026 Shortlisted and awarded at the Tsinghua School of Architecture Doctoral Forum for four consecutive years
- 2025.12 NeurIPS 2025 (Spotlight, Poster) — [San Diego, USA](#)
- 2024.09 IFLA 2024 (Oral) — [Istanbul, Türkiye](#)
- 2024.05 ICASSP 2024 (Poster) — [Seoul, Korea](#); 9th Youth Geography Forum (Poster) — [Xiamen, China](#)
- 2023.11 Territorial Spatial Planning Annual Conference — First Prize, [Guangzhou, China](#)

DESIGN & COMPETITION AWARDS

- 2026.05 Tencent Location-Based Services Developer Contest — [Second Prize](#)
- 2026.04 51st International Exhibition of Inventions of Geneva — [Silver Medal](#)
- 2025.05 18th Challenge Cup Guangdong Student Academic Science & Tech Competition — [Grand Prize](#)
- 2024.05 Milan Design Week University Exhibition — [First Prize \(Guangdong & National\)](#)
- 2023.06 2nd National Community Garden Design & Build Competition — [First Prize](#)
- 2023.04 Garden Expo Student Design Competition — [Gold](#); IFLA Asia-Pac LA Awards — [Honorable Mention](#)

EXPERIENCE

- 2026.06–2026.08 **Research Intern (LLM Agents)**, iFlytek
- 2023.10–2024.01 **Data Analysis Intern**, Innovation Center, Guangzhou Urban Planning & Design Survey Research Institute
— *analyzed 400M+ raw mobile-signaling records.*

PATENTS

All patents are first- or second-inventor (advisor first).

- 2026.01 Cultural-heritage perceived-value analysis & retrieval system based on multi-agent LLMs
- 2025.02 Method for park social-media review scoring and key-pharse extraction
- 2025.02 Park image focal-object extraction based on visual saliency prediction
- 2024.01 Bimodal scoring method coupling park social-media review text and images (**granted**)
- 2023.12 Built-environment perception assessment based on street-view images